Compressor Rotor Joints and Interference Fit Study

Mechanical Engineering Senior Project

By:
Aubrey Perez
Isabela Jimenez
Isaiah Guerrero
Jireh Velasquez

Senior Project
Sponsor: *Solar Turbines*
California Polytechnic
University, San Luis Obispo
College of Engineering

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Contents of this Report

This report fulfills the Senor Project requirement for the Mechanical Engineering Department. This document contains:

1. Scope of Work (SOW)

2. Preliminary Design Review (PDR) Report

3. Critical Design Review (CDR) Report

4. Final Design Review (FDR) Report

5. Project Expo Poster

Compressor Rotor Joints and Interference Fit Study

Scope Of Work (SOW)

By:
Jireh Velasquez
Isaiah Guerrero
Aubrey Perez
Isabela Jimenez

Senior Project
Sponsor: Manuel Aldana (Solar Turbines)
California Polytechnic University,
San Luis Obispo
College of Engineering

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Project Overview	**Key Specifications**
Solar Turbines need a comprehensive study on the effects axial and radial inference fits have on turbine compressor rotor stiffness. The aft weldment of the compressor has press-fit dowel pins and pilot joints to connect one weldment section to another. These fits need to be assessed through a finite element modal analysis, as well as practically tested through modal impacts to investigate how the fits affect the stiffness of the turbine's connected weldments. The experimental data will be used to validate the finite element analysis (FEA) model.	The engineering specifications listed in the table below are made to define how we will achieve Solar Turbines' requests during this project. Each specification is made to be validated through either a specified target or completion.

Specification	Target Value
Frequency	> 186 Hz
Effective Stress	< 20 MPa
Heat	1400 °C
Modal Analysis	Completion
Iterations	3 Iterations
Physical Tolerance	>±0.0001''
Machining	Completion
Weight	50 lbs.

Scope and Deliverables

- *CAD model of the rotor part we are working on*
- *Researched material properties*
- *CAD model of compressor assembly including multiple parts as needed*
- *ANSYS model to assist in analyzing stiffness and natural frequency*
- *Completed model test procedure on a physical part*
- *Final results of natural frequency and stiffness of parts*
- *A final report summarizing all findings*

What's Next

The next step of our project is to provide a full preliminary design review (PDR) of our plan to conduct the press fit analysis study. Within the next four weeks, we will ideate action plans and use various decision matrices to finalize our testing procedure. From there, we will proceed with the testing plan and gather results to present in the critical design review (CDR) and final design review (FDR).

Overview

Solar Turbines is a company that specializes in creating turbine engines with the purpose of power generation for prolonged periods of time. The turbine engines are designed to run for ~30,000 hours non-stop before requiring a complete engine overhaul. Some phenomena affecting engine vibration that Solar Turbines is experiencing with their engines are shifting between the aft (AFT) weldment and forward (FWD) weldment, resulting in reduced stiffness of the compressor rotor. Due to these issues, Solar Turbines has asked us to investigate these issues and propose a potential solution. Our main objective is to analyze how different axial and radial interference fits affect the stiffness of the compressor. The goal is to reduce shifting between weldments. The stiffness in rotor system comes from two sources; the rotor geometry and the bearings. The rotor geometry provides stiffness but the joints at the curvics or the weldments introduce flexibility (hence reducing the stiffness). For us to determine if our calculations would increase rotor stiffness, we will require experimental data from Solar Turbines regarding their current FWD and AFT weldment and compare those values to our experimental data. Overall, the scope of our project is to increase rotor stiffness and reduce shifting between weldments. To ensure that we are well informed and produce a product that not only meets these requirements, but surpasses expectations, we researched topics regarding interference fits and rotor dynamic stiffness. Moreover, we meticulously analyzed our customer needs and engineering specifications to ensure our design is compatible with current production and meets requirements.

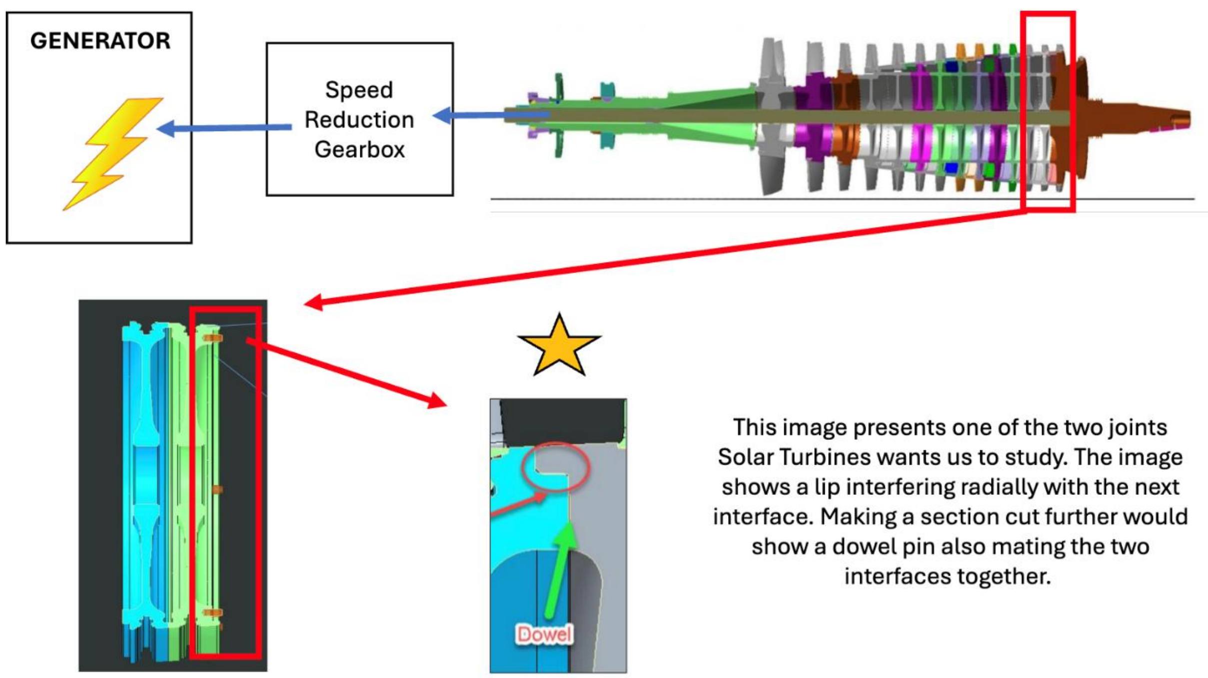


Figure 2. Boundary sketch illustrating project scope

Researched Topics and Existing Solutions

The main type of research we conducted for this project was technical research, as our project focuses on technical testing and analysis for Solar Turbines. Rather than having various other products to compare with, we researched the fit connections used by competing turbine companies. Our goal was to prepare ourselves for considering other mating connections for rotors and set up the testing plan for the prototype modal testing.

Table 1: Existing Solutions

This table details the different types of fits previously used to address issues most similar to that of Solar Turbines and provides examples of their usage.

Solution Type	Description	Examples
Turbine shroud with two axial pin mounting and cover plates	This solution consists of a turbine shroud assembly where ceramic shroud segments are attached to a metal carrier using axial pins and sealed cover plates to handle high temperatures and thermal expansion. The pins transfer loads and maintain alignment while allowing some movement for thermal expansion. The sealed cover plates close and seal the holes preventing the pins from ejecting and blocking gas leaks. Some improvements include reducing the number of parts and optimizing pin geometry.	[1]
Turbine shroud with spring-loaded axial pins	In this solution, the metal carrier and the ceramic blade track segment are connected by an axial pin and retainer plug. The retainer plug locks the pin in place. A “biasing” element or spring is placed between the head of the axial pin and carrier to keep the components pressed against each other. Many flanges are used to help with alignment and support for the blade track segment. Some drawbacks of this solution are that the tolerances for manufacturing are very high to ensure the correct preload of the springs, and the added complexity makes it difficult to assemble.	[2]
Finite Element Method – Based Interference Fit Design	This approach uses finite element analysis to model the three-dimensional stresses and deformations in interference-fitted mechanical assemblies. Unlike traditional methods based on Lame’s equations, FEM accounts for non-linear contact behavior, friction, and complex geometries, resulting in more accurate predictions of stress, deformation, and safety factors. The study introduces two safety factors to guide design tolerance and improve reliability: one for structural strength and one for torque transmission to guide design tolerance and improve reliability.	[3], [4], [5]
Bolts and flanges	In this solution, the casing sections are joined with circumferential bolted flanges and horizontally split joints that keep the compressor and turbine aligned while allowing easy maintenance. These connections provide stiffness, sealing, and precise alignment between rotating and stationary components under high temperature and pressure. This relates to our project because, like interference and dowel-pin fits, these joints rely on controlled contact and preload to maintain structural	[6]

	integrity and stiffness under cyclic loading. This solution could be improved by strategically placing the bolts to make the turbine easier to assemble and disassemble.	
Tapered interference fit	This solution works by pressing two matching conical surfaces together to create frictional resistance through elastic deformation and contact pressure. They provide strong, reliable joints with high holding strength and resistance to loosening, especially when small taper angles and sufficient contact length are used. Their performance can be improved by optimizing taper geometry and surface friction to maximize efficiency while avoiding plastic deformation during assembly.	[7], [8], [9]
Curvic couplings	Curvic couplings work by using interlocking curved teeth on mating faces to precisely align and rigidly join rotating components, transmitting torque while maintaining accurate concentricity. They perform well in providing repeatable assembly, high load capacity, and excellent alignment in turbine and compressor rotors. They can be improved by refining tooth geometry to reduce stress concentrations, applying advanced coatings or materials for better wear resistance, and using modern manufacturing techniques to achieve tighter tolerances and smoother tooth surfaces.	[10], [11], [12]

Customer Needs

Our main customer, Solar Turbines, requested a stiffness analysis on their turbine rotor according to the press and pilot fits used to hold AFT and FWD weldment rotors together. Rather than having a list of vague needs/ wants, we are provided with a list of goals to reach over the course of the project. The customers' needs listed below are similar to the ones that can be seen in our QFD in Appendix A.

Table 2: Customer Needs

This table lists the identified customer requirements with justification and assigns a relevant, measurable engineering specification to be used to evaluate the adequacy of the prototype in addressing this need.

Customer Needs	Relevant Engineering Specifications (Spec #)	Justification
Prevent shifting	Spec1: Frequency	We currently assume natural frequency will directly relate to the amount of shifting seen through the spring constant.
ANSYS model	Spec 2: Heat Consideration Spec 3: Stress Analysis Spec 4: Modal Analysis	The ANSYS model(s) developed must include considerations for heat transfer, stress, and natural frequencies as these are all important parameters in the system being studied.
Stiffness vs. Interferences	Spec 5: Multiple iterations	Multiple iterations will allow us to deliver different parts that the customer can choose from for their needs.
Machinable	Spec 6: Tolerances	The part we are studying calls for tight tolerances. Tight tolerances increase the difficulty in machining.
Able to Assemble	Spec 7: Assembly/ Disassembly Time	The part must not be overly complex to where technicians need intense training and time to complete task.
Maintain Production Cost	Spec 8: Machining	Machining will be the most expensive part of our project. Tight tolerances, long machining time, jig preparation and other factors must be heavily taken into consideration.
Physical Prototype Model	Spec 9: Weight	The part should not be overly heavy, but strong enough to hit stress specifications and factor of safeties.

Engineering Specifications

The engineering specifications table is a list of achievable goals meant to describe how we will achieve the end goal of providing Solar Turbines with a stiffness analysis on their AFT and FWD weldment rotors. These specifications are made to be specific thresholds on technical achievements while having a “compliance” column to explain how we plan to measure the achievements. A rough draft of these specifications can be seen in the QFD in Appendix A.

Table 4: Engineering Specifications

This table lists the identified engineering specifications and assigns a specific target value or ability to be used to evaluate the adequacy of the prototype in addressing each need.

Spec #	Spec	Specification Description	Target	Tolerance	Risk *	Compliance **
1	*Frequency*	Need to have a natural= frequency greater than 186hz	*>186 hz*	*+inf, -0.0*	*H*	*A, T*
2	*Heat*	New mating method must function at operating temperature	1400 C	±200	*M*	*A*
3	*Stress*	Lower stress caused by interference fits while maintaining sufficient mating strength	<20Mpa	±1 MPa	*L*	*T*
4	Modal	Create modal analysis model of the compressor assembly	complete	[-]	L	I
5	Iterations	Create a minimum of 3 different mating configurations	complete	[-]	L	A,T
6	Tolerance	Create tolerances that are machinable within reason	>0.0001”	±0.001	H	T
7	Assembly/ disassembly	Create mating configurations that can be assembled and disassembled with minimal tooling	complete	[-]	L	I
8	Machining	Create mating surfaces that are machinable without complex and unique tooling	complete	[-]	H	I

9	Weight	Prototype weldment should be a reasonable weight that could be easily transported by a person	50 lb.	±5 lb.	L	A, I

[5] S. P. Singh and R. K. Sharma, “Interference Fit Simulation on Pin Joint by Using ANSYS,” *International Journal of Research in Engineering and Technology (IJRET)*, vol. 4, no. 5, May 2015. (ijret.org) * Risk of meeting specification: (H) High, (M) Medium, (L) Low

** Compliance Methods: (A) Analysis, (I) Inspection, (S) Similar to Existing, (T) Test

Additional Background

In the turbine system, mass primarily comes from the disks, forward cone, aft hub, and shafts. The overall stiffness is determined by the rotor geometry, though joints such as weldments and curvic couplings introduce flexibility that reduces stiffness; bearings contribute additional stiffness to the system. Damping is mainly provided by an oil wedge in the bearings. At operating speeds up to 120,000 rpm (for the T130 turbine), the rotor exhibits several mode shapes, including banana and parabola modes, with small deflections of 1–3 thousandths of an inch. Most Solar Turbines machines use tilt-pad radial and thrust bearings, and rotor dynamic analyses model the system with simplified axi-symmetric and revolved components; blades are represented as point masses and bearings are modeled using hydrodynamic film theory. Critical modes for the T130 are around 2400 rpm (turbine overhung), 4500 rpm (compressor “banana” mode), and 9300 rpm (“spaghetti” mode near full speed). Depending on the rotor’s length-to-diameter ratio, either single-plane or two-plane balancing is used to improve dynamic imbalance and vibration. Sensors, such as proximity probes and accelerometers, are used to measure displacement, casing vibration, and overall dynamic behavior.

Annotated Bibliography

[1] J. Freeman, A. D. Sippel, and D. J. Thomas, “Turbine Shroud Assembly With Sealed Pin Mounting Arrangement,” Apr. 30, 2025. Accessed: Oct. 23, 2025. [Online]. Available: https://patentimages.storage.googleapis.com/30/92/a2/d49c5a0007e201/EP4095355B1.pdf

This patent describes a turbine shroud assembly for a gas turbine engine, featuring a seal segment supported by a carrier and a mount system that uses pins and a cover plate to secure the seal while managing thermal-expansion stresses. It is relevant to our senior project because it offers a practical mechanical solution for high temperature sealing in turbine systems, which is directly applicable to the thermal management portion of our work.

[2] T. J. Freeman, A. D. Sippel, and D. J. Thomas, “Turbine Shroud Assembly With Axially Biased Pin And Shroud Segment,” Apr. 30, 2025 Accessed: Oct. 23, 2025. [Online]. Available: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://patentimages.storage.googleapis.com/30/92/a2/d49c5a0007e201/EP4095355B1.pdf

This patent describes a turbine shroud assembly designed for gas turbine engines, involving a carrier segment, shroud segment, and an axially biased pin/biasing member arrangement to maintain sealing engagement under thermal expansion. It is relevant to our project as it addresses maintaining tight seals and structural integrity under high temperature/thermal-cycling conditions, which is helpful for thermal conditions we are studying. From this document we learned how a biasing spring can apply preload to a shroud segment via a pin-mount arrangement, improving sealing at the gas path interface despite thermal expansion.

[3] Zhang, Y., et al. “Design of Interference Fits via Finite Element Method.” *International Journal of Mechanical Sciences*, vol. 42, no. 9, 13 June 2000, pp. 1835–1850. *ScienceDirect*.

This article applies the finite element method to model interference fits and shows that traditional thick-wall cylinder theory can be inadequate. It is relevant because our project involves high-precision mechanical joints subject to thermal expansion and structural loads, and this work gives insight into how FEM improves fit-design accuracy. From the paper, we learned about the effects of three-dimensional stress distributions in interference fits and how design adjustments influence contact pressures and long-term reliability.

[4] Karmankar, Rahul G. “Stress Analysis of Interference Fit by FEM.” *International Research Journal of Engineering and Technology*, vol. 02, no. 09, Dec. 2015, pp. 2641–2648.

This document presents a finite element analysis of interference fits using ANSYS to evaluate contact and Von Mises stresses for various allowances between a pin and hole. It is relevant to our turbine project because it demonstrates how FEA can be applied to study stress distribution and contact behavior in precision-fit mechanical assemblies. We learned how changes in interference allowance affect stress levels, which helps us better understand how to optimize fits and avoid material failure in turbine components.

[5] Imran, Mohammed, et al. “Interference Fit Simulation on Pin Joint by Using Ansys.” *International Journal of Research in Engineering and Technology*, vol. 04, no. 05, May 2015, pp. 579–583.

This document presents an ANSYS-based finite element simulation of an interference fit in a pin and plate joint using steel and aluminum materials. It is relevant to our turbine project because it demonstrates how FEA can predict stress distribution, deformation, and contact behavior in precision mechanical joints. We learned how different materials respond to interference fits, helping us better understand how material choice and fit tolerances affect stress performance and joint reliability.

[6] R. C. Stancliff, "The General Electric LM5000 Marine Gas Turbine," in *Proceedings of the ASME Turbo Expo*, Toronto, ON, Canada, Jun. 4–8, 1989, Paper No. 89-GT-13. [Online]. Available:https://asmedigitalcollection.asme.org/GT/proceedings/GT1989/79146/V002T03A001/238533.

This paper describes the General Electric LM5000 marine gas turbine, a high-power engine adapted from an aircraft design for marine applications. It is relevant to us because it demonstrates how precise casing joints and connections maintain stiffness, alignment, and sealing under high thermal and mechanical loads. We learned how bolted flanges are designed to manage stress and ensure reliability.

[7] D. Bozkaya and S. Muftu, "Efficiency considerations for the purely tapered interference fit (TIF) abutments used in dental implants," *Journal of Biomechanical Engineering*, vol. 126, no. 4, pp. 393–401, Aug. 2004, doi: 10.1115/1.1784473.

This paper discussed how tapered interference fits create strong mechanical joints by using elastic deformation and frictional pressure between conical surfaces, offering high stability and resistance to loosening. This insight relates to your turbine project because similar principles can improve the reliability of tapered shaft or hub connections—optimizing taper angle and surface finish can enhance holding strength while preventing material yielding during assembly or operation.

[8] D. Bozkaya and S. Muftu, "Mechanics of the tapered interference fit in dental implants," *Journal of Biomechanics*, vol. 36, no. 11, pp. 1649–1658, 2003, doi: 10.1016/S0021-9290(03)00177-5.

This paper discussed how tapered interference fits create strong, self-locking joints by generating frictional resistance through elastic or partially plastic deformation of the contacting conical surfaces. The study showed how taper angle, contact length, and interference depth control pull-out strength and stress distribution, with excessive interference leading to material yielding. This relates to our project because similar principles apply to press-fitted or tapered shaft–hub connections. Optimizing taper geometry and fit pressure can improve joint reliability while avoiding overstress or slip under dynamic loading.

[9] N. Vishwakarma, A. Renuke, and V. M. Phalle, "Effect of tapered interference fit between impeller and shaft in turbo machines," *Journal of Mechanical Engineering – Strojnícky časopis*, vol. 68, no. 3, pp. 25–32, 2018, doi: 10.2478/scjme-2018-0024.

This journal article discusses how using a tapered interference fit between the impeller and shaft reduces stress concentrations and maintains strong contact pressure compared to a conventional cylindrical fit. This design improves torque transmission and allows for higher rotational speeds without the need for keys or splines. It relates to our project because similar principles can be applied to improve joint reliability, reduce fatigue at connection points, and ensure stable operation under high-speed rotational loads.

[10] “Back to Basics: Curvic Coupling Design,” *Gear Technology*, pp. 34–48, Nov.–Dec. 1986, Gleason Works, Rochester, NY.

This paper discussed how curvic couplings use precision-ground curved teeth to align and lock rotating components together, ensuring accurate centering and efficient torque transmission. They are highly effective in maintaining rigidity and repeatability in multi-disk assemblies like turbine rotors. This relates to our project because the same principles of precise alignment, torque transfer, and structural integrity are critical for connecting rotor sections and ensuring smooth, reliable turbine operation.

[11] S. R. Pisani and J. J. Rencis, “Investigating CURVIC coupling behavior by utilizing two- and three-dimensional boundary and finite element methods,” Engineering Analysis with Boundary Elements, vol. 24, no. 4, pp. 271–275, 2000.

This article discusses how curvic couplings effectively transmit torque and maintain precise alignment between rotating components while highlighting the importance of analyzing stress concentrations at the tooth roots. The study showed that accurate 3D modeling is essential to predict these stresses and improve component life. It is relevant to our project because we will face similar challenges in ensuring reliable torque transfer, minimizing stress concentrations, and maintaining alignment in large rotating structures.

[12] X. Jiang, Y. Zhu, J. Hong, and Y. Zhang, “Development and validation of analytical model for stiffness analysis of curvic coupling in tightening,” *Journal of Aerospace Engineering*, vol. 27, no. 4, p. 04014012, 2014, doi: 10.1061/(ASCE)AS.1943-5525.0000348.

This article discusses how the stiffness of a curvic coupling is strongly influenced by its geometry, friction characteristics, and the bolt tightening process, which together affect how torque and loads are transferred between rotating components. The study showed that optimizing parameters like tooth height and pressure angle can stabilize stiffness and reduce stress concentrations. This relates to our project because similar coupling behavior and load transfer mechanisms are critical for ensuring structural stability, precise alignment, and reliable torque transmission in large rotating assemblies.

Appendix A: QFD House of Quality

The QFD house of quality is an expansive diagram showcasing the design considerations and specifications we will try to achieve during this project. The diagram considers Solar Turbines as the main customer, and some of Solar Turbines' competitors as the "current products". As this diagram is often used for product creation, we used the "customer needs/ wants" section as a list of tasks required of us by Solar Turbines. This diagram is what was used to make the engineering specifications seen in the Scope of Work.

Correlations	
Positive	+
Negative	–
No Correlation	

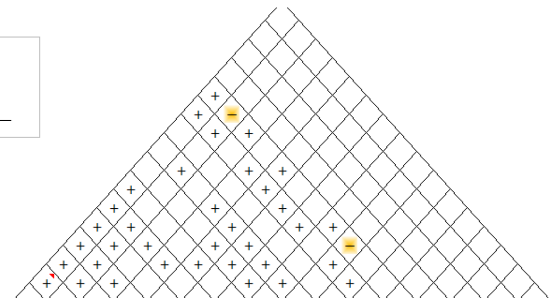
Relationships	
Strong	●
Moderate	○
Weak	▽

Direction of Improvement	
Maximize	▲
Target	◇
Minimize	▼

QFD House of Quality

Project: CCRP

Revision Date: 10/16/25



Row #	Weight Chart	Relative Weight	Solar Turbines	Solar Turbine Costumers	Machinists/ Assembly		Maximum Relationship	WHAT: Customer Requirements (Needs/Wants) / HOW: Engineering Specifications (Tests)	Create a CAD model of the rotor	Determine standard material properties	Create CAD model of compressor assembly	Create Ansys model	Create prototype model	Modal impact Ansys model	Modal impact prototype	Research of alternative mating methods	Lower natural frequency range	Create SOW, PDR, CDR	Verifiy experimental data with calculated data	Determine manufacturing process					General Electric	Siemens Energy	Rolls Royce			Row #
								Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16						
	WHO: Customers							Direction of Improvement	▲	▲	◇	▲	▲	◇	◇	▲	▼	◇	▲	▼					NOW: Curr. Products					
1	IIIIIIII	16%	8	10	3		9	Prevention of Shifting	●	●	●	○	●		▽	○	▽	▽	●	●					3	3	3			1
2	II	6%	10	1	1		9	Ansys Model	●	●	○	●	▽	●	▽	○	●		●	▽					1	2	1			2
3	IIII	9%	10	1	5		9	Whether Radial or Axial Fits Have Greater Effect on Stiffness	●	●	○	●	●	●		○	○	▽	●	○					1	1	1			3
4	IIII	9%	10	1	5		9	Increase assembly stiffness	●	●	●	●	●	●	○	○	○		●	●					4	3	3			4
5	IIIII	12%	8	1	10		9	Design for manufacturing	●	●	○	▽	○	▽		○	▽	○	▽	●					4	4	4			5
6	IIIIIIII	17%	10	10	3		9	Increase Assembly life	○	●	●	○	▽	●	●	●	○		○	○					4	3	4			6
7	IIIII	12%	10	5	3		9	Reduce production cost	▽	○	○	○	●		▽	●	●	○	○	●					3	3	4			7
8	IIIIII	12%	10	1	10		9	Create CAD model	●	○	○	○	●	●		○	●	○	▽	○					1	1	1			8
9	III	8%	8	1	5		9	Create Protoype Model	●	●	●	▽	●		●	○	●	○	●	○					1	1	1			9
10		0%																												10
11		0%																												11
12		0%																												12
13		0%																												13
14		0%																												14
15		0%																												15
16		0%																												16

	HOW MUCH: Target Values	Target for Spec #1	Target for Spec #2	Target for Spec #3	Target for Spec #4	Target for Spec #5	Target for Spec #6	Target for Spec #7	Target for Spec #8	Target for Spec #9	Target for Spec #10	Target for Spec #11	Target for Spec #12				
	Max Relationship	9	9	9	9	9	9	9	9	9	3	9	9				
	Technical Importance Rating	705.8	755.9	598.8	400.5	648.2	485.7	283.8	471.1	470.9	155.8	536.1	577.2	0	0	0	0
	Relative Weight	12%	12%	10%	7%	11%	8%	5%	8%	8%	3%	9%	9%	0%	0%	0%	0%
Curr. Products	General Electric	4	3	4	2	2	2	3	3	4	0	4	4				
	Siemens Energy	4	4	3	1	1	3	3	3	2	0	3	4				
	Rolls Royce	3	4	3	2	2	3	4	2	3	0	4	3				
	0																
	0																
	Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Appendix B: Non-Cited Research

This appendix includes the citations for other research we completed but were not cited in this specific report. The other sources we found to help us with our general research are listed below.

[13] Bonisoli, Elvio, et al. “Interference Fit Estimation through Stress-Stiffening Effect on Dynamics.” *Mechanical Systems and Signal Processing*, vol. 160, 16 Apr. 2021. *ScienceDirect*.

[14] Sikanen, Eerik, et al. *Effect of Operational Temperature on Contact Dynamics of Shrink-Fitted Compressor Impeller Joint*. Cham Springer, 14 June 2019, pp. 3341–3351.

[15] Erena, Diego, et al. “NUMERICAL STUDY on the INFLUENCE of ARTIFICIAL INTERNAL STRESS RELIEF GROOVE on FRETTING FATIGUE in a SHRINKFITTED ASSEMBLY.” *Depósito de Investigación de La Universidad de Sevilla* , vol. 151, Nov. 2020.

[16] Leedham, L. H., and M. I. Mech.E. “Some Problems in the Manufacture of Experimental Gas Turbines.” *Proceedings of the Institution of Mechanical Engineers*, vol. 163, no. 1, Nov. 1950.